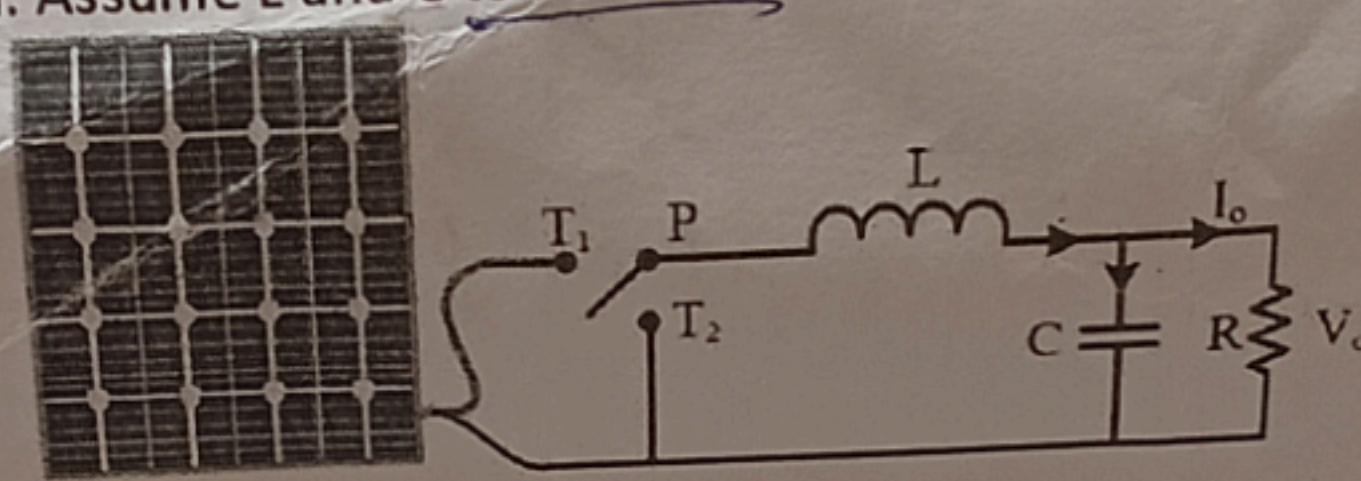


Department of Electrical Engineering
B. Tech. III Semester Mini Examination August 2025
 Sub: Measurement & Instrumentation (EE-24214)

Time: 50 Minutes

Max. Marks: 20

Note: Solve all questions else do as directed. Assume any missing data with due justification.
 Approximate to two decimal. Calculators are allowed.

		COs	Marks
Q.1	A 500W solar PV based system is shown in the Fig.1. The output DC voltage of the solar panel is 325V and it is connected to a buck converter. The node (P) voltage of the buck converter is 48V. The engineer has two basic D'arsonval meter in his possession. He has to measure the output voltage (V_o) of the buck converter and the output current (I_o) using these available meters. The full scale deflection current of the meter is 1mA and internal resistance is 20000Ω for both the meters. Design a complete measuring system such that D'arsonval meter1 should operate as a voltmeter and D'arsonval meter2 as an ammeter. The range of designed meter should be 1.5 times the quantity to be measured. Assume L and C to be ideal.  Solar Panel Fig.1	CO2	5
Q.2	For the system designed in Q1, during measurement of current, what will be the amount of current flowing through the coil of the D'arsonval meter configured as ammeter? In the meter, the coil has 100 turns, with length of the coil (l) as 1.5cm and width (d) as 1cm. Calculate the deflecting torque on the coil. Assume the magnetic flux density is 0.2T. If the spring constant is 1.89×10^{-6} Nm/rad at the equilibrium, what will be the angular deflection in degree?	CO2	4
Q.3	Due to external disturbance there is a 5V noise embedded in the output voltage being measured in the designed voltmeter in Q1. Determine the signal to noise ratio at the output.	CO2	2
Q.4	Later the engineer decides to purchase a new voltmeter from the market with sensitivity of $2\Omega/V$, instead of the voltmeter designed in the Q1. The range of the voltmeter purchased is 0-100V. What is the output voltage indicated by the new voltmeter. Also calculate the error in percentage. Assume L to be 1mH and C to be ideal. The operating frequency is 10kHz.	CO2	4
Q5	List down the different types of static characteristics used in measuring instruments? Also classify them based on desirable and undesirable characteristics.	CO1	5

Course Coordinator: Dr. T. S. Bheemraj
 Date of Examination: 22nd August 2025

$N = 100$
 $l = 1.5 \text{ cm}$

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